

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

The efficient approximation of intractable posterior distributions is a central problem in Bayesian inference. Stein Variational Gradient Descent (SVGD) confronts this problem as a deterministic variational method that transports an ensemble of interacting particles toward a target distribution. Because the SVGD dynamics unfold within a reproducing kernel Hilbert space (RKHS), their behavior is decisively shaped by the kernel, which simultaneously governs the rate of convergence and the fidelity of the resulting approximation. The median heuristic, though the de facto standard for setting the kernel bandwidth, is a single-scalar rule that cannot adapt to the geometry of the target and, as we document, degrades markedly as the dimension grows. We break with this fixed-kernel practice and instead select kernel parameters adaptively over an arbitrary parameterized family of kernels. Recent convergence analyses framed in terms of the kernelized Stein discrepancy (KSD) identify maximization of the KSD over the kernel as precisely the criterion that accelerates the underlying dynamics; building on this principle, we introduce Adaptive SVGD (Ad-SVG), which interleaves SVGD particle transport with gradient ascent on the KSD with respect to the kernel parameters and thereby tunes the kernel online at negligible additional cost. We develop a convergence theory that lifts existing guarantees for fixed-kernel KSD minimization to the adaptive regime, establishing that the maximal KSD over the kernel class vanishes along the iterates and, under a Stein log-Sobolev inequality, quantifying convergence in KL divergence together with explicit iteration-complexity and bias bounds. These guarantees are borne out empirically: across Gaussian mixtures, an ODE-governed linear inverse problem, high-dimensional Gaussian and Gaussian-process targets, and Bayesian logistic regression, Ad-SVG consistently surpasses the median heuristic and, in particular, remedies the variance collapse that afflicts SVGD as the dimension increases.

1 Introduction

Stein Variational Gradient Descent (SVGD) [SVGD] is a deterministic, particle-based method for approximate Bayesian inference that has established itself as a compelling alternative to Markov Chain Monte Carlo (MCMC). SVGD evolves an ensemble of interacting particles along update directions given by the functional gradient of the Kullback–Leibler (KL) divergence to the target, with the admissible directions constrained to the unit ball of a reproducing kernel Hilbert space (RKHS). This construction ties the method’s efficacy directly to the geometry of that space, and therefore to the kernel: the kernel governs the pairwise interaction between particles and, through it, both the rate of convergence and the quality of the approximation [Duncan2019OnTG, Nsken2021SteinVG]. A well-documented symptom of a poorly chosen kernel is that the particle approximation systematically underestimates the posterior variance [DBLP:conf/iclr/BaEGSSWZ22]. Observations of this kind have hardened into a widely held belief that SVGD is inherently ill-suited to high-dimensional targets. We argue, to the contrary, that this apparent limitation is largely

an artifact of a fixed, hand-tuned kernel rather than a defect of the dynamics themselves. To substantiate this claim, we introduce an adaptive mechanism that tunes the kernel parameters online during inference by maximizing the kernelized Stein discrepancy (KSD), thereby restoring effective particle transport in complex, high-dimensional spaces and reframing kernel selection as the decisive determinant of SVGD’s performance.

1.1 Related work.

Since its introduction [SVGD], SVGD has become a widely used tool for approximate Bayesian inference across a broad spectrum of machine learning applications [DBLP:conf/uai/LiuRPL17, DBLP:conf/iclr/MessaoudMXP0CC24, DBLP:conf/nips/PuGHLHC17, DBLP:journals/tsp/Kassabandits2019] and its theoretical foundations have been clarified substantially in parallel. Mean-field convergence has been characterized in both the continuous-time [LLN2019, Duncan2019OnTG, NEURIPS2020’16f8e136] and discrete-time [NEURIPS20203202111c, pmlr-v162-salim22a] regimes; finite-particle convergence rates have been established under a range of assumptions [balasubramanian2024improvedfiniteparticle, NEURIPS2023’54e5d7af]; and the dynamics have been identified with gradient flows on spaces of probability measures [DBLP:conf/nips/Liu17, Duncan2019OnTG], in direct analogy with the variational formulation of VariationalFormulationFokker.

A single factor runs through this entire body of work and proves decisive: the kernel. Because it dictates the pairwise interaction between particles, the kernel determines both the trajectory and the limiting behavior of the method (see also Figures 1 and 9). The available convergence results are typically cast as mean-field convergence in KSD, yet the extent to which KSD convergence entails weak convergence is itself governed by the kernel [pmlr-v70-gorham17a]. Against this backdrop, the ubiquitous median heuristic [JMLR:v13:gretton12a] is expedient but theoretically unjustified, and its performance is known to deteriorate systematically as the dimension grows [DBLP:conf/iclr/BaEGSSWZ22, DBLP:conf/icml/ZhuoLSZCZ18, DBLP:conf/icml/WangZ01]. Efforts to counteract this degradation in high dimensions have proceeded along two largely orthogonal lines: modifying the transport dynamics themselves [stein’newton, NEURIPS2019’5dcd0ddd, gong2021sliced, pmlr-v151-liu22a] and learning the update direction with neural networks [langosco2022neural, ZHAO2023118975]. Both introduce additional architectural machinery, and both thereby leave unanswered the more basic question we take up here: how much can be recovered by kernel selection alone.

Closest to our work is ai’stein’2023, which introduces a mixture-of-kernels framework: a KSD is defined for a weighted linear combination of base kernels, and the mixture weights are learned by maximizing this multiple-kernel KSD. That construction is nonetheless confined to a finite kernel basis and does not address the continuous parameter optimization we develop here. Our formulation subsumes their setting as a special case and extends the maximization principle to arbitrary parameterized kernel families, including the dimension-dependent bandwidths that, as our experiments show, prove essential in high dimensions.

1.2 Contributions.

This work is organized around a single, deliberately isolated research question: can SVGD, stripped of any architectural or algorithmic augmentation, attain competitive performance solely by being equipped with a principled, adaptively chosen kernel? By decoupling kernel selection from the many other modifications that populate the SVGD literature, we can attribute the resulting performance

gains unambiguously to the kernel and quantify precisely how far adaptive kernel learning alone extends the reach of the original method. Our findings answer this question in the affirmative and, in doing so, recast kernel selection from an incidental hyperparameter choice into the primary lever governing the behavior of SVGD in high dimensions.

Our contributions are the following.

- (i) **Adaptive kernel selection.** We introduce Ad-SVGd, which tunes the kernel online by ascending the KSD between the current particle measure and the target throughout inference. In contrast to the median heuristic, which commits to a single scalar bandwidth, our formulation optimizes an arbitrary vector of continuous kernel parameters—in particular, per-dimension bandwidths—and applies to any parameterized kernel family, all at no additional gradient-evaluation cost over vanilla SVGD.
- (ii) **Convergence theory for the adaptive regime.** We analyze Ad-SVGd in the discrete-time mean-field setting and lift the fixed-kernel convergence guarantees of prior work to a kernel that is re-selected at every step. We establish that the maximal KSD over the parameterized class converges to zero as the particle distribution approaches the target, both for exact and for approximate KSD maximization. Under a Stein logarithmic Sobolev inequality—which, notably, we require of only a single kernel in the class—we further obtain convergence in KL divergence, together with explicit iteration-complexity rates and a bias bound for the case of fixed-accuracy inner maximization.
- (iii) **Empirical validation across dimensions and tasks.** On Gaussian mixtures, an ODE-governed linear inverse problem, high-dimensional Gaussian and Gaussian-process targets, and Bayesian logistic regression, Ad-SVGd consistently outperforms the median heuristic and, decisively, remedies the variance collapse that degrades SVGD as the dimension increases.

2 Mathematical background

We fix throughout a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ with associated RKHS $\mathcal{H}_0$, and we take $\mathcal{H}$ to be the d -fold Cartesian product of $\mathcal{H}_0$ equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$; the canonical feature map is $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ denotes its subspace of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is defined by

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ denotes the set of couplings of μ and ν , that is, the probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [IPM] quantify the distance between two measures as the maximal discrepancy of integrals taken over a prescribed class of test functions. To render this construction tractable when measuring the deviation of a sample from an intractable target, Stein’s

method `[bounderrornormal]` supplies test functions with zero mean under the target: for the Stein operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ `[KernelTestGoodness, KernelizedSteinDiscrepancy]`. Restricting the supremum to the unit ball of $\mathcal{H}$ then yields the *kernelized Stein discrepancy* (KSD),

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. This variational problem admits the closed-form maximizer $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$, where $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ and $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, with Φ_k the feature map of the kernel k . The supremum therefore coincides with the RKHS norm of ψ , which yields the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$. This identity is precisely what renders the discrepancy computable and, as we exploit below, differentiable with respect to the kernel parameters.

2.2 Stein Variational Gradient Descent.

Given a target π and a reference measure μ_0 , SVGD transports μ_0 toward π through the iteration $\mu_{n+1} := T_\# \mu_n$, in which $T_\#$ is the push-forward under the map $T(x) = x + \gamma \psi^{\mu_n}(x)$ and the vector field $\psi^\mu = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, $\mu \in \mathcal{P}(\mathbb{R}^d)$, is the direction of steepest descent of the KL divergence. The choice of ψ^{μ_n} is far from incidental: the maximizer of Equation (1) establishes that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is precisely the element of the unit ball of $\mathcal{H}$ that minimizes $\frac{d}{d\gamma} \text{KL}(T_\# \mu_n \| \pi) \Big|_{\gamma=0}$ `[SVGD]`, with $\text{KL}(\cdot \| \cdot)$ the KL divergence. Along this optimal direction the instantaneous rate of KL decrease equals exactly the squared discrepancy,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_\# \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying the maps T constructed in this manner iteratively to a particle set $\{X_0^i\}_{i=1}^M$ drawn from μ_0 yields the empirical particle updates

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because establishing convergence of SVGD directly in KL divergence is difficult and hinges on restrictive assumptions, recent analyses have instead measured convergence through the kernelized Stein discrepancy `[NEURIPS20203202111c, NEURIPS2023'54e5d7af, pmlr-v162-salim22a]`. Convergence in KSD is not, however, self-sufficient: it does not by itself imply weak convergence, since the sequence of mean-field measures may fail to be tight—a necessary condition for weak convergence in Polish spaces—and, as the variational problem Equation (1) makes explicit, the RKHS structure, and hence the kernel, dictates how faithfully KSD convergence transfers to weak convergence `[pmlr-v70-gorham17a]`. The kernel is therefore not a mere tuning knob but the very object that determines what convergence in KSD actually certifies—which is precisely why we elevate it to the target of adaptation.

To exploit this dependence rather than merely suffer it, we formulate our adaptive variant of SVGD over a parameterized family of kernels $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$ we write KSD_θ for the corresponding kernelized Stein discrepancy, ψ_θ^μ for the induced optimal update direction, and Φ_θ for the associated feature map. Rendering the kernel dependence explicit in this way allows us to select θ adaptively throughout inference, so that the kernel’s influence on convergence is actively harnessed and the well-documented fragility of a single fixed kernel is eliminated.

The guiding principle of our approach is to select, at each step, the kernel that maximizes the KSD between the current empirical particle distribution and the target. The rationale is direct and quantitative: under SVGD the instantaneous decrease in KL divergence equals the squared KSD at the current particle measure (Equations (2) and (7)), so maximizing the KSD over the kernel class is precisely maximizing the per-step rate of KL decrease. Geometric analyses of SVGD as a gradient flow corroborate this criterion, establishing that kernels inducing larger KSD values yield strictly more favorable convergence behavior when the associated RKHSs are compared [Nskn2021SteinVG, Duncan2019OnTG]. Maximizing the KSD is therefore not a heuristic surrogate but the natural optimality criterion dictated by the SVGD dynamics themselves.

Whereas vanilla SVGD seeks convergence of the KSD for a single fixed kernel, our algorithm targets the largest KSD attainable within the entire kernel class. In contrast to the median heuristic of **SVGD**, which is tied to one specific isotropic kernel, the procedure applies to any parameterized family: at every step it seeks an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

Concretely, before each particle update we adjust the kernel parameter by one or more gradient-ascent steps on the KSD with step size $s > 0$, and—to control runtime—we optionally refresh the parameter only intermittently rather than at every SVGD step (Algorithm 1), a choice governed by the user-specified flag `paramupdate`. The design is deliberately modular: the base transport step may be replaced by more sophisticated SVGD variants (adaptive step-size schedules, line searches, momentum, or second-order and preconditioned updates), and the gradient ascent on θ by any alternative optimizer, without altering the kernel-selection mechanism in either case. Our concrete kernel update rests on the closed-form representation [**KernelTestGoodness**, **KernelizedSteinDiscrepancy**]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

from which the gradient with respect to θ follows by direct differentiation. Two observations establish that this adaptivity is obtained at essentially no additional cost relative to vanilla SVGD:

- The kernel-parameter ascent reuses the very gradients $\nabla \log \pi(X_n^i)$ already computed for the SVGD transport step; Ad-SVGd therefore requires no additional evaluations of $\nabla \log \pi$, which are typically the dominant expense.
- The residual overhead is reduced further by taking only a small number of inner steps `nstepstheta`, or by refreshing the kernel parameter only once per several SVGD steps; and

when the ensemble size M is large, a subsample of particles suffices to approximate the KSD. In our experiments, these controls render the runtime of Ad-SVGD indistinguishable from that of the median heuristic.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps$, $nstepstheta \in \mathbb{N}$
Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to  $nsteps - 1$  do
    Decide  $paramupdate \in \{\text{True}, \text{False}\}$ ;
    if  $paramupdate$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $nstepstheta - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To place Ad-SVGD on firm theoretical ground, we extend the recent KSD-based convergence theory of SVGD to the adaptive setting. Specifically, we build on the analysis of [pmlr-v162-salim22a], which operates under Talagrand’s inequality—a condition that holds under mild assumptions on the target and is strictly weaker than the logarithmic Sobolev inequality routinely invoked elsewhere. Extending that analysis to a kernel that varies across iterations requires only that the assumptions of [pmlr-v162-salim22a] be imposed uniformly over all kernel parameters, which we now state.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand's inequality

Throughout we write

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

for the KL functional and rely on the following descent inequality. Fix a kernel parameter $\theta \in \Theta$ for which Assumption 3 holds, and define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for an arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, for $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ satisfying the step-size restriction $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, Proposition 3.1 of [pmlr-v162-salim22a] yields

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. Specializing this bound to the iterates (3) for a fixed kernel parameter $\theta \in \Theta$ yields the descent condition of [pmlr-v162-salim22a], namely

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The crux of (7) is the verification of the step-size condition $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, which Talagrand's inequality (Assumption 2) secures and which in turn licenses (6) with $g = \psi_\theta^{\mu_n}$. Summing (7) then gives $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ with $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$, whence

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0.$$

The essential observation—and the one that makes the adaptive extension immediate—is that this argument is entirely local in the kernel: under an adaptive parameter sequence (θ_n) , the descent condition (7) carries over verbatim with θ replaced by the current θ_n ,

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

We first treat the idealized case in which the kernel maximization is solved exactly, so that the adaptive iteration assumes the form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

that is,

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

This exact formulation is available whenever the KSD admits a (unique) closed-form maximizer over Θ . Such is the case for the multiple-kernel SVGD framework of [ai'stein'2023], in which the kernel is a convex combination of base kernels,

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1,$$

so that our exact analysis subsumes that method as a particular instance. In the general setting the maximizer θ_n in (9) must be approximated, and we assume throughout that $\arg\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

For this exact scheme, applying the fixed-kernel descent condition at each step with the maximizing parameter yields the following guarantee, an immediate consequence of Theorem 3.2 in [pmlr-v162-salim22a].

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) = 0$.*

The more realistic—and more broadly applicable—situation is that in which the KSD maximization is solved only approximately, as is the case for the gradient-ascent inner loop of Algorithm 1. Assuming $\max_{\theta \in \Theta} \text{KSD}_\theta(\mu | \pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$, we analyze the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

for a sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ tasked with maximizing $\text{KSD}_\theta(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient-ascent update on the KSD; but—consistent with the modularity emphasized above—any iterative optimizer exhibiting the same asymptotic behavior may be substituted without affecting the analysis. That behavior is captured by a single, mild requirement on the optimization error.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this assumption alone, the maximal KSD over the entire kernel class is driven to zero.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n|\pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [pmlr-v162-salim22a] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n|\pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n|\pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq \text{KSD}_{\theta_n}^2(\mu_n|\pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

This guarantee can be sharpened considerably. Under a Stein logarithmic Sobolev inequality, we upgrade the convergence of Ad-SVGd from the KSD to the KL divergence itself; for concrete decay profiles of ε_n we obtain explicit iteration-complexity rates for the mean-field system (10), and the same setting furnishes an exact characterization of the bias incurred when the KSD maximization is solved only to a fixed accuracy $\bar{\varepsilon}$ per iteration. The full development is deferred to Appendix B. A distinctive and practically consequential feature of the adaptive setting is that the log-Sobolev condition need hold for only *one* kernel in the class $\{k_{\theta}\}$ rather than for a prescribed kernel, which affords substantial flexibility in verifying it in practice.

5 Numerical Experiments

We now evaluate Ad-SVGd empirically, structuring the study so as to isolate the effect of kernel adaptation across regimes of increasing difficulty. The main text reports two complementary experiments: a one-dimensional Gaussian mixture, which exposes the acute sensitivity of SVGd to the bandwidth, and a linear inverse problem governed by an ordinary differential equation, which tests the method on a genuinely multivariate posterior. Experiments probing the effect of growing dimensionality on Gaussian and Gaussian-process targets are collected in Appendix A, and Section 5.4 establishes the advantage of Ad-SVGd over the median heuristic on the Bayesian logistic regression benchmark standard in the SVGd literature. Throughout, the comparison is deliberately controlled: Ad-SVGd and median-heuristic SVGd share the same transport dynamics, step sizes, and particle counts, so that any difference in approximation quality is attributable to kernel selection alone.

5.1 Kernel parameterization

SVGD is most commonly instantiated with kernels of the form

$$k_h(x, y) = \exp \left(-\frac{\|x - y\|_p^p}{h} \right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [SVGD, DBLP:conf/iclr/BaEGSSWZ22, Duncan2019OnTG] and the object of selection is the scalar *bandwidth* h . The standard heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, with med the current median pairwise distance between particles—a rule motivated by balancing the two terms of the SVGD update (3) [SVGD]. Its single scalar necessarily collapses all coordinate directions onto one common length scale, and it is precisely this rigidity that our adaptive scheme is designed to remove.

To realize the full flexibility of the adaptive method, we adopt anisotropic product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp \left(-\frac{|x_i - y_i|^p}{h_i} \right)$$

parameterized by a bandwidth vector $h = (h_1, \dots, h_d)$, thereby assigning an independent length scale to each coordinate. The derivatives required to instantiate Algorithm 1 for this family (via Equations (4) and (5)) are available in closed form. For a controlled comparison we additionally evaluated a per-dimension variant of the median heuristic that assigns each coordinate its own naive median; this variant failed to match the adaptive scheme and exhibited pronounced variance collapse, confirming that the gains stem from KSD-driven selection rather than from anisotropy alone.

The same construction points directly to more scalable kernel families for higher dimensions. Spectral or low-rank parameterizations can adapt to the dominant directions of variability without incurring the cost of a full product kernel, while Matérn or mixed-product constructions afford additional control over smoothness and anisotropy; each fits directly into the parameterized-kernel formulation of Section 3 and the accompanying convergence theory, without modification.

5.2 Toy example

We begin with the one-dimensional benchmark of **SVGD**, the two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. Initializing the particles from $\mathcal{N}(0, 1)$ with $M \in \{50, 200, 500\}$, we run SVGD for 10^4 steps at step size 1, using the kernel above with $p = 1$ across a range of fixed bandwidths h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see **StatisticalAspectsWasserstein**) against an exact reference sample of size 10^5 . Figure 1 reports the outcome and renders the central difficulty concrete: SVGD attains a small $\mathcal{W}_1$ only

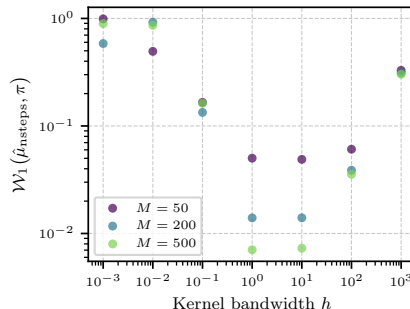


Figure 1: Final Wasserstein 1-distance $\mathcal{W}_1$ to the target for the one-dimensional Gaussian mixture, obtained by SVGD with a range of fixed bandwidths h (lower is better). Performance is competitive only within a narrow band of h and degrades sharply on either side of it, exposing the acute bandwidth sensitivity that motivates adaptive selection.

for bandwidths lying within a narrow interval, with accuracy deteriorating steeply on either side. Even in one dimension, then, performance is governed almost entirely by h , so that a principled selection strategy is not merely helpful but decisive—and it is exactly this gap that Ad-SVGD is designed to close.

5.3 Linear inverse problem based on ODE

We now turn to a genuinely multivariate posterior arising from a partial-observation inverse problem, adapted from [Weissmann2022Optimization]. Consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

together with the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete, noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ denotes observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is given by the composition $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ the observation operator evaluating f at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we place a Gaussian process (GP) prior on u through the truncated Karhunen–Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients are independent, $x_k \sim \mathcal{N}(0, \lambda_k)$ with $\lambda_k = 50k^{-2}$. Inference therefore reduces to estimating the KL coefficient vector $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under the prior $\mathcal{N}(0, \Gamma_0)$, $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. With Gaussian observation noise $\varepsilon \sim \mathcal{N}(0, \Gamma)$, Bayes' rule furnishes the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation we replace H by a numerical discretization of Equation (11) on a grid of mesh size 2^{-8} and consider the fully observed system ($N_{\text{obs}} = 2^8$). We retain $N_x = 16$ KL terms for u and set the noise covariance to $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare SVGD with the median heuristic—henceforth Med-SVGD—against Ad-SVGD across ensemble sizes $M \in \{50, 100, 200\}$, using the kernels of Section 5.1 with $p = 1$. Both methods are run for $4 \cdot 10^5$ iterations at particle step size 10^{-3} under the AdaGrad-based step-size control

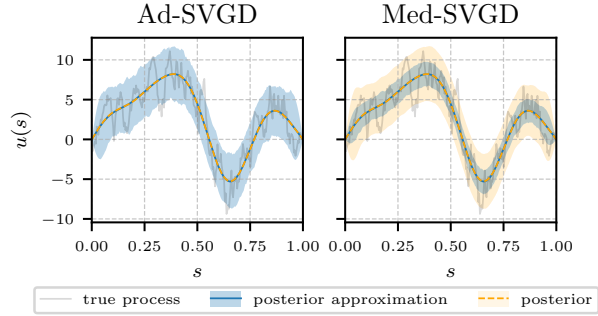


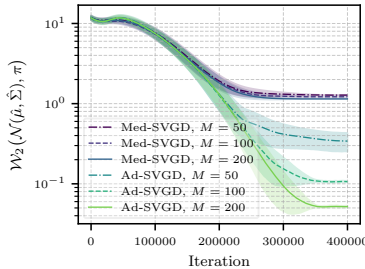
Figure 2: Gaussian-process reconstruction for the ODE-based inverse problem (mean and 90% confidence interval). Both methods recover the posterior mean, but only Ad-SVGD reproduces the posterior uncertainty band, whereas median-heuristic SVGD collapses

recommended by **SVGD**; Ad-SVGD updates the bandwidth vector with step size 10^{-5} once per 100 particle updates. Under this schedule the two methods are computationally indistinguishable, so that the comparison isolates the effect of kernel adaptation at matched cost.

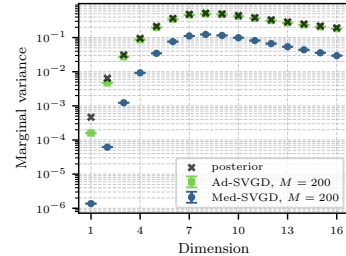
Figure 2 displays the GP reconstruction for a representative seed: both methods recover the posterior mean accurately, but only Ad-SVGD reproduces the posterior uncertainty, whereas Med-SVGD collapses it. To quantify approximation quality we adopt the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ fitted to the particle sample mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [StatisticalAspectsWasserstein]. Because π is multivariate normal, this admits the explicit form

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

with μ_π, Σ_π the posterior mean and covariance; we additionally compare the marginal variances of the final particles against those of the posterior. Figure 3 aggregates these results over 56 random seeds (the posterior covariance being independent of $\bar{y}$). Ad-SVGD attains a uniformly smaller $\mathcal{W}_2$ error than Med-SVGD, which systematically underestimates the posterior uncertainty; moreover, and in sharp contrast to Med-SVGD, the accuracy of Ad-SVGD continues to improve as the ensemble grows beyond 50 particles—direct evidence that adaptation enables the method to exploit additional particles rather than saturate. Figure 4 traces the bandwidths selected by Ad-SVGD: the component-wise bandwidths stabilize appreciably faster than the approximation error, and the converged values differ markedly across coordinates, being negatively correlated with the corresponding marginal variances. The method thus discovers an interpretable, anisotropic length scale that a single median bandwidth is fundamentally unable to represent.



(a) Convergence of the $\mathcal{W}_2$ approximation error

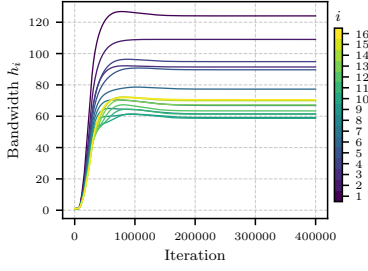


(b) Marginal variances vs. the posterior

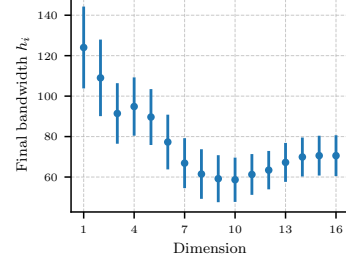
Figure 3: ODE-based inverse problem: Med-SVGD versus Ad-SVGD (mean and 95% confidence interval over 56 random seeds). Ad-SVGD attains a smaller $\mathcal{W}_2$ error and recovers the marginal variances, whereas Med-SVGD systematically underestimates them; the Ad-SVGD error, moreover, continues to improve as the ensemble size grows.

5.4 Bayesian logistic regression

Following **SVGD** and **pmlr-v151-liu22a**, we evaluate the methods on a Bayesian logistic regression (BLR) model for the Covertypes data set [covertypes], a standard SVGD benchmark. The regression weights w receive a hierarchical Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and the target of inference is the joint posterior of $x = [w, \alpha]$. As in **pmlr-v151-liu22a**,



(a) Per-dimension bandwidth evolution (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Bandwidths selected by Ad-SVGD for the ODE-based inverse problem, aggregated over 56 random seeds. The component-wise bandwidths stabilize faster than the approximation error and settle at clearly distinct, anisotropic values that are negatively correlated with the marginal variances.

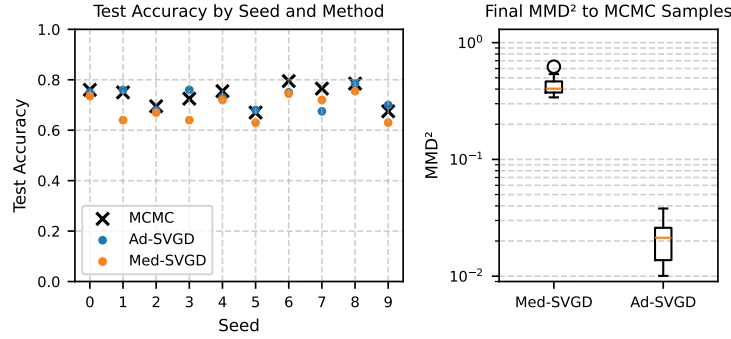


Figure 5: Bayesian logistic regression: approximation quality of the SVGD particles, measured by test prediction accuracy (left) and squared maximum mean discrepancy MMD^2 to the NUTS reference (right), over 10 random draws. Across nearly all seeds, Ad-SVGD attains both higher accuracy and a smaller MMD^2 than Med-SVGD.

we obtain a high-quality reference posterior from the No-U-Turn Sampler (NUTS) of **nuts**, against which the approximation quality of Med-SVGD and Ad-SVGD is assessed. Following **pmlr-v151-liu22a**, we test on random subsets of size 1000 and aggregate over 10 draws, reporting the classification accuracy on a held-out test set obtained from the particle mean. As Figure 5 shows, Ad-SVGD dominates Med-SVGD across nearly all seeds, simultaneously achieving higher test accuracy and closer agreement with the MCMC reference—so that the improved posterior fidelity translates directly into improved predictive performance.

As a direct measure of posterior fidelity, the same figure reports the squared maximum mean discrepancy (MMD^2) [**MMDref**] to the MCMC samples, on which the two methods differ markedly. Figure 7 corroborates this finding: the final-particle covariance matrices, aver-

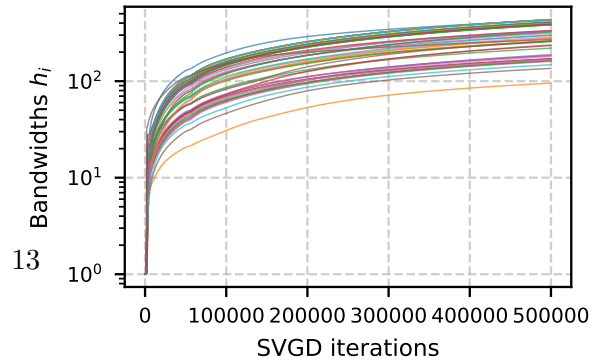


Figure 6: Bandwidth evolution for the BLR example.

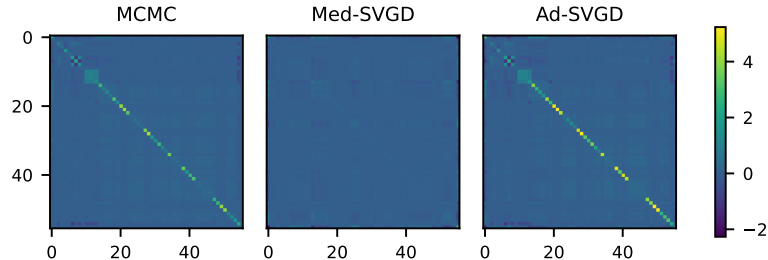


Figure 7: Covariance matrices of the final SVGD particle sets compared to the MCMC reference (mean over 10 seeds). Med-SVGD collapses the off-diagonal structure and underestimates the variances, whereas Ad-SVGD closely reproduces the reference covariance.

aged over the 10 seeds (following the visualization of [pmlr-v151-liu22a](#)), reveal that Med-SVGD severely underestimates the posterior uncertainty, while Ad-SVGD reproduces it far more faithfully. Figure 6 exposes the underlying mechanism: starting from a uniform, uninformed initialization (1 in every component), Ad-SVGD adaptively drives the component bandwidths across several orders of magnitude to values matched to the posterior geometry—an adjustment entirely beyond the reach of any single scalar bandwidth.

6 Limitations and outlook

The principal limitation of our theory is its reliance on Assumption 4, which posits that the inner optimizer eventually closes the gap to the maximal KSD. Establishing precisely when the alternating gradient-ascent scheme of Algorithm 1 provably satisfies this condition is the subject of ongoing work; while we do not yet certify it in full generality, the strong and consistent empirical performance across every experiment we report indicates that it is met throughout the regimes of practical interest. We emphasize that this assumption concerns only the inner kernel optimization, and not the transport dynamics, whose convergence we then establish unconditionally.

Because our analysis targets the original SVGD dynamics, the adaptive kernel-selection mechanism is by design orthogonal to modifications of the transport step and composes with them directly. Combining it with recent variants such as sliced SVGD [[gong2021sliced](#)], Grassmann SVGD [[pmlr-v151-liu22a](#)], or Stein transport [[nsken2024steintransportbayesianinference](#)] is therefore a natural and especially promising avenue, holding out the prospect of compounding the gains from adaptive kernels with those from improved dynamics.

A More numerical experiments

A.1 Gaussian mixture models

We first return to the mixture of Section 5.2. To render the bandwidth sensitivity of Figure 1 tangible at the level of the distribution, Figure 9 overlays the final particle histograms on the target density for $h \in \{0.001, 1, 1000\}$, illustrating how markedly the fixed-bandwidth approximation deteriorates once it departs from the well-tuned regime. We then compare Med-SVGD and Ad-SVGD directly, matching the final particle distributions after 10^4 iterations at step size 1 across a range of ensemble sizes M .

Figure 8 reports the $\mathcal{W}_1$ -distance between the final empirical distribution and the target for $M = 10, 20, 50, 100, 200, 500$. As expected, accuracy improves monotonically with M for both methods, and both reach $\mathcal{W}_1 < 0.01$ at $M = 500$. In this benign one-dimensional setting adaptive selection is therefore on par with a well-tuned median heuristic—a finding that precisely localizes the substantial gains reported elsewhere: they emerge specifically where the fixed heuristic breaks down, namely in higher dimensions, rather than reflecting an indiscriminate advantage.

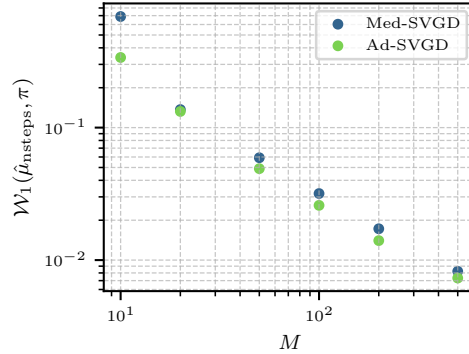


Figure 8: Approximation error ($\mathcal{W}_1$) for the 1D Gaussian mixture as a function of the particle count. Med-SVGD and Ad-SVGD perform comparably in this benign regime, both attaining $\mathcal{W}_1 < 0.01$ at $M = 500$.

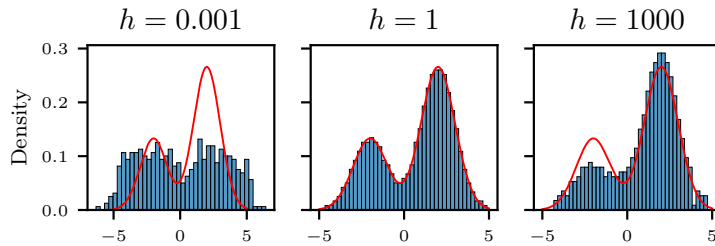


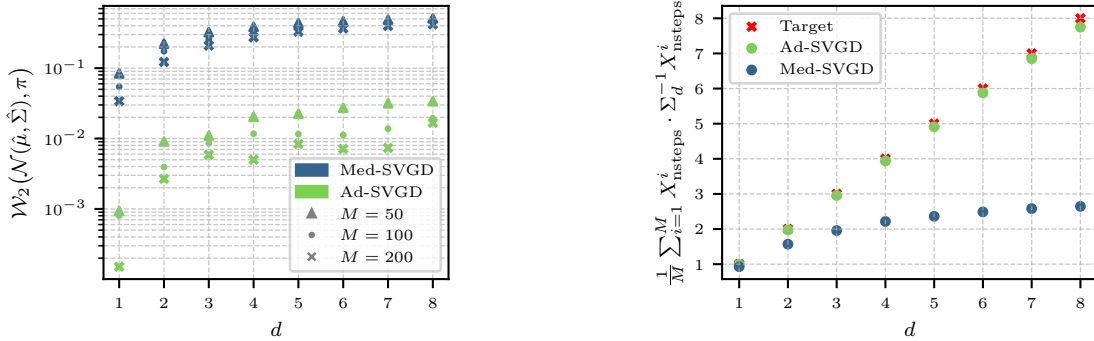
Figure 9: Histograms of the final particle set for fixed bandwidths $h = 0.001, 1, 1000$, with the target density π in red. Only the intermediate bandwidth recovers the bimodal target; the extreme values either collapse the particles or oversmooth them, rendering visible the sensitivity quantified in Figure 1.

In one dimension, both methods approximate the target well: adaptive bandwidth selection performs on par with the median heuristic without conferring a decisive advantage in this regime. This parity is informative rather than negative—it delineates precisely the boundary of the median heuristic’s competence and sets the stage for the higher-dimensional experiments below, where that competence fails and adaptation becomes essential.

A.2 Scaling dimension: multivariate normal distribution

To probe the high-dimensional regime systematically, we consider the anisotropic Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$, whose sharply differing coordinate scales are precisely what a single scalar bandwidth cannot accommodate. The particles are initialized from $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and both algorithms are run for 10^4 iterations at step size 0.1, reducing the step size and lengthening the run whenever numerical stability so requires.

We track the approximation quality of the final particle set for Med-SVGD and Ad-SVGD as d grows. Since the $\mathcal{W}_2$ -distance between Gaussians is available in closed form (see [StatisticalAspectsWasserstein]) we adopt as our quality measure the $\mathcal{W}_2$ -distance between π_d and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ fitted to the empirical particle mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [StatisticalAspectsWasserstein]; following the convention of the Python Optimal Transport library [PythonOptimalTransport] used in our computation, we refer to this quantity as the *Bures–Wasserstein distance*. Figure 10a plots this measure for $M = 50, 100, 200$ as a function of dimension, and the separation is unambiguous: Ad-SVGD substantially outperforms Med-SVGD at every value of d .



(a) Bures–Wasserstein error for $M \in \{50, 100, 200\}$. (b) χ^2 test statistic for $M = 200$ (target value d).

Figure 10: Multivariate normal targets of increasing dimension. Ad-SVGD attains a smaller Bures–Wasserstein error at every d (a), and its χ^2 statistic tracks the ideal value d , whereas that of Med-SVGD diverges as d grows (b)—direct evidence of the progressive variance underestimation incurred by the median heuristic.

The Bures–Wasserstein distance is relatively insensitive to the small variances in the trailing components of π_d , so to expose how well each method captures uncertainty in exactly those directions we turn to the test statistic $X \cdot \Sigma_d^{-1} X$, which for $X \sim \mathcal{N}(0, \Sigma_d)$ is χ^2 -distributed with d degrees of freedom and hence has expected value d (cf. [Metropolisadjustedinteracting]). Evaluating its mean over the $M = 200$ particles and comparing to the ideal value d in Figure 10b yields a clean diagnostic: the Med-SVGD statistic drifts progressively further below d as the dimension grows, whereas the Ad-SVGD statistic remains close to d throughout. Med-SVGD therefore fails to represent the higher-dimensional targets, while Ad-SVGD continues to capture them accurately. Tables 1 and 2 render this quantitative at the level of individual coordinates, listing for each d the marginal variances of the particle set alongside the target variances (first row) for Med-SVGD and Ad-SVGD respectively. Ad-SVGD recovers the target marginal variances accurately across all components and all dimensions, whereas Med-SVGD substantially underestimates them: in the leading component the Med-SVGD variance falls from 0.93 at $d = 1$ to 0.48 at $d = 8$, while Ad-SVGD

Table 1: Marginal variances of the final Med-SVGD particle distribution ($M = 200$) for the d -dimensional Gaussian targets; the target marginal variances π_d appear in the first row. Med-SVGD underestimates the variances across all components, and the shortfall grows systematically as d increases.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final Ad-SVGD particle distribution ($M = 200$) for the d -dimensional Gaussian targets; the target marginal variances π_d appear in the first row. Ad-SVGD recovers the target variances closely in every component and at every dimension.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

holds near 0.97–1.00—in full agreement with the χ^2 diagnostic of Figure 10b.

Finally, we examine the marginal particle distributions for $M = 200$ and $d = 8$ in detail. Normalizing each coordinate by the inverse of the corresponding target marginal standard deviation (equivalently, scaling the i -th component by i) renders every marginal of π_d a standard normal. The resulting histograms are shown in Figures 11 and 12 for Med-SVGD and Ad-SVGD, each against the standard Gaussian density. Ad-SVGD reproduces all eight marginals faithfully, whereas Med-SVGD visibly understates their spread; the Q–Q plots of Figures 13 and 14, which compare the normalized marginal quantiles against the standard-normal quantiles, convey the same conclusion component by component.

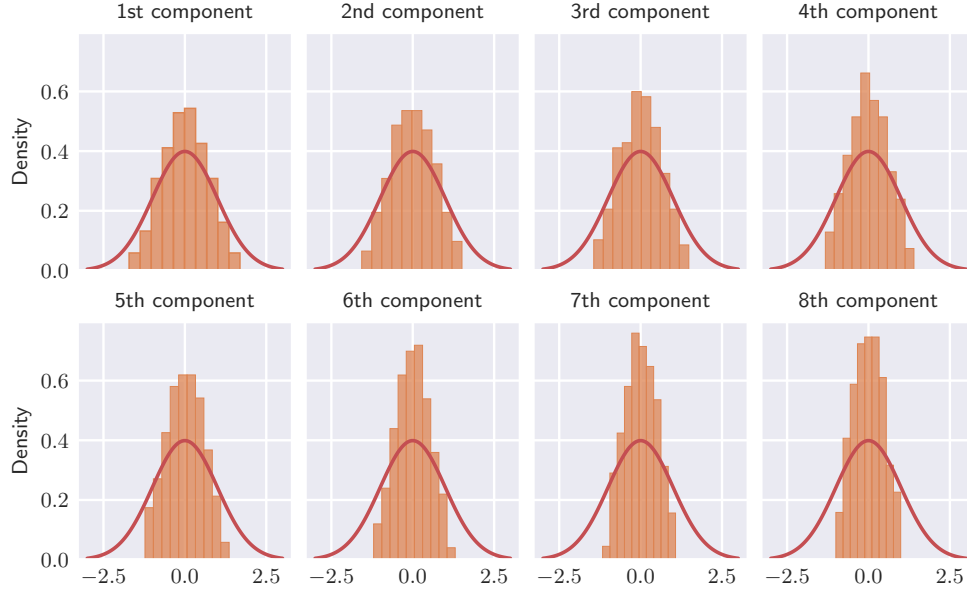


Figure 11: Component-wise histograms of the final Med-SVG particles for the eight-dimensional target ($M = 200$), rescaled by the marginal target standard deviations so that each target marginal is standard normal (shown for comparison). The particle marginals are visibly overconcentrated, confirming the systematic variance underestimation.

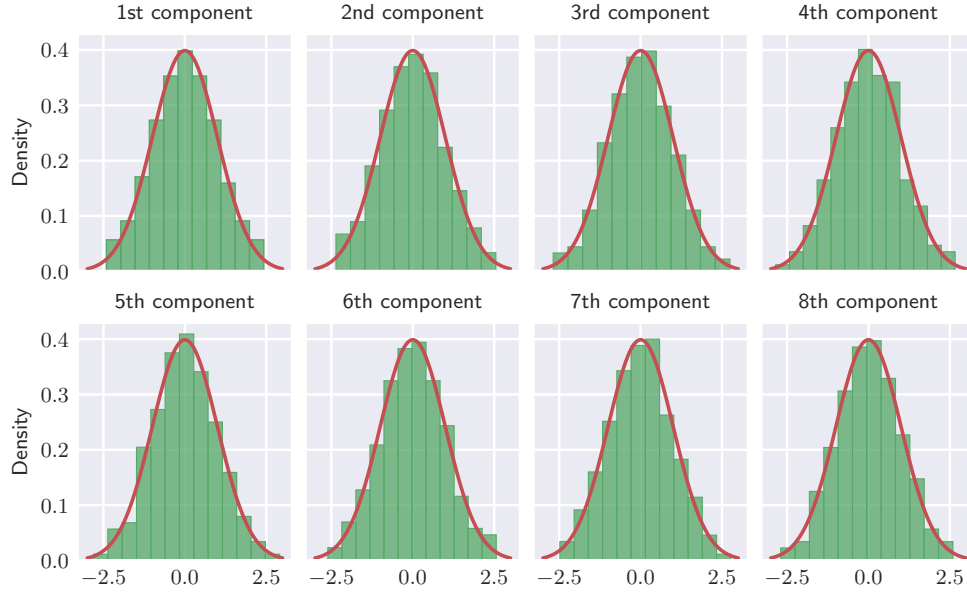


Figure 12: Component-wise histograms of the final Ad-SVG particles for the eight-dimensional target ($M = 200$), rescaled by the marginal target standard deviations so that each target marginal is standard normal (shown for comparison). All eight marginals are recovered accurately.

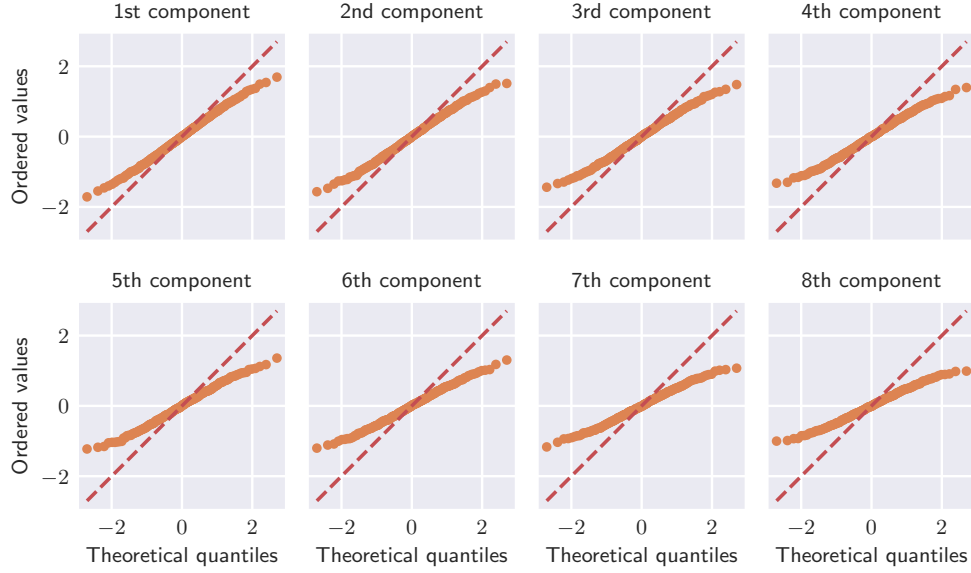


Figure 13: Q–Q plots of the normalized marginals of the final Med-SVGD particles for the eight-dimensional target ($M = 200$) against the standard normal, with the identity line of unit slope through the origin shown for reference. The departures from the identity line directly reflect the underestimated marginal spread.

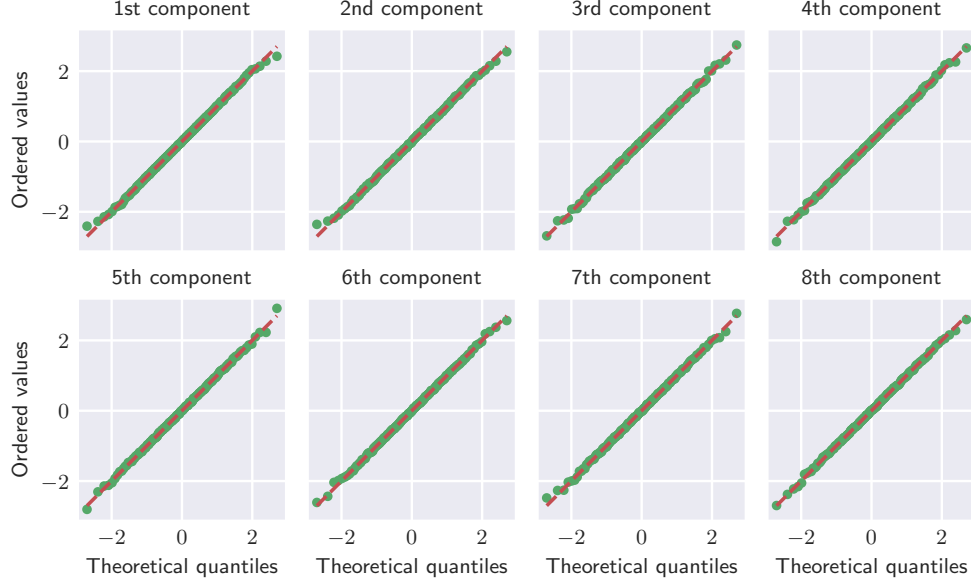


Figure 14: Q–Q plots of the normalized marginals of the final Ad-SVGD particles for the eight-dimensional target ($M = 200$) against the standard normal, with the identity line of unit slope through the origin shown for reference. The marginal quantiles align closely with the identity line across all eight components.

A.3 Gaussian process inference

As a further high-dimensional test, we study the Gaussian-process (GP) inference task of [reich21], which allows us to vary the latent dimension and the number of observations independently. We consider a GP on $[0, 1]$ represented by a truncated Karhunen–Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ collects independent Gaussian coefficients $x_k \sim \mathcal{N}(0, k^{-2})$. Observing the process at N_y equispaced points in $[0, 1]$ and inferring the coefficients x_k yields, for fixed N_x and N_y , a linear-Gaussian inverse problem with forward model $Y = AX + \varepsilon$, in which the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with Σ diagonal and entries k^{-2} , $k = 1, \dots, N_x$, and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are constructed by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

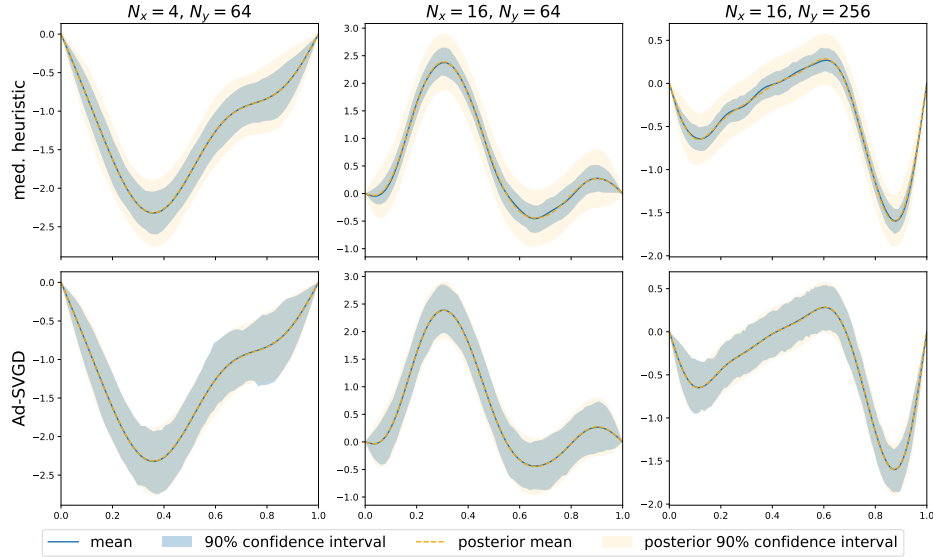


Figure 15: Estimated processes from median-heuristic SVGD and Ad-SVGD ($M = 100$ particles) for three model configurations, compared to the posterior. As the latent dimension grows, the median heuristic collapses the posterior uncertainty band, whereas Ad-SVGD tracks it faithfully; the effect is consistent across the number of observations.

We sample from the resulting posterior $X | Y = \bar{y}$ with SVGD and again contrast the median heuristic with the adaptive approach. Median-heuristic SVGD uses the isotropic kernel $k(x, y) = \exp(-\|x - y\|_1/h)$, whereas Ad-SVGD uses the anisotropic product kernel $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with a distinct bandwidth h_i per coordinate; for the $N_x = 16$ configurations we employ the AdaGrad-based step-size control of [SVGd].

Figure 15 contrasts the estimated processes with the posterior for the configurations $(N_x, N_y) = (4, 64)$, $(16, 64)$, and $(16, 256)$. As the latent dimension increases, median-heuristic SVGD progressively underestimates the posterior variance, whereas Ad-SVGD sustains an accurate approximation; and because this pattern persists across the different observation counts, the advantage is attributable to the latent dimension rather than to the amount of data. Table 3 quantifies the comparison through the trace of the particle covariance against the true posterior: the median heuristic severely understates the total variance, while Ad-SVGD recovers it closely in every configuration.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the final particle covariance ($M = 100$ particles) against the theoretical posterior trace across latent dimensions N_x and observation counts N_y , averaged over 25 runs. Ad-SVGD closely matches the theoretical trace in every configuration, whereas the median heuristic recovers less than half of it in each case.

B Refined theoretical analysis

We now strengthen the convergence result under a Stein log-Sobolev inequality, which couples the KL divergence directly to the KSD. This inequality was introduced by **Duncan2019OnTG**, who also discussed conditions under which it holds. A distinctive benefit of the adaptive setting is that we can substantially weaken this requirement: rather than demanding it of a prescribed kernel, we ask only that *some* kernel in the class satisfy it, since the maximization over θ automatically selects the most favorable member. This is made precise in the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property allows us to quantify convergence directly in KL divergence. Notably, it delivers an explicit error bound on $\mathcal{F}(\mu_n)$ even under the merely approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, that is, without requiring the inner optimization error to vanish—so that a fixed-accuracy kernel maximization already suffices to control the bias.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□